

nlp-practical-1

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NLP - Practical 1

Perform tokenization (Whitespace, Punctuation-based, Treebank, Tweet, MWE) using NLTK library. Use porter stemmer and snowball stemmer for stemming. Use any technique for lemmatization. Input / Dataset –use any sample sentence

```
[2]: # =====
# NLP PREPROCESSING USING NLTK
# =====

import nltk
import string
from nltk.tokenize import (
    word_tokenize,
    WhitespaceTokenizer,
    TreebankWordTokenizer,
    TweetTokenizer,
    MWETokenizer
)
from nltk.stem import PorterStemmer, SnowballStemmer
from nltk.stem import WordNetLemmatizer
```

```
[14]: # -----
# DOWNLOAD REQUIRED DATA (Run once)
# -----

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
```

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[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!

```

[14]: True

```

[10]: # -----
# SAMPLE INPUT TEXT
# -----
text = """The 52-hertz whale is one of the most fascinating and mysterious_
↳creatures ever detected in the oceans, often
referred to as the "loneliest whale in the world." This whale was first_
↳identified in 1989 by researchers from the Woods Hole Oceanographic
Institution while analyzing underwater acoustic data collected by the U.S. Navy.
↳ What made this whale extraordinary was the frequency of its
call-52 hertz-which is much higher than the typical vocalizations of other_
↳large whale species. For example, blue whale usually communicate at
frequencies between 10 and 39 hertz, while fin whale calls are typically_
↳around 20 hertz. Because sound travels differently underwater, whales
rely heavily on low-frequency calls to communicate across vast distances,_
↳sometimes hundreds of kilometers. The 52-hertz whale's unusual pitch
suggests that other whales may not be able to hear or recognize its calls,
which is why it has been popularly described as "lonely," though this_
↳interpretation is more poetic than scientifically confirmed.
Over the years, the whale has been tracked moving across the North Pacific_
↳Ocean, following migration patterns somewhat similar to other
baleen whales, yet always singing at its unique frequency. Scientists are still_
↳unsure what species it belongs to; some hypothesize it
could be a hybrid between different whale species, while others think it might_
↳be an individual with a physical anomaly that affects its
vocalization system. Despite its uniqueness, the whale does not appear to be_
↳distressed-it continues to travel, feed, and vocalize consistently,
suggesting it is surviving normally. The idea of its loneliness has captured_
↳global imagination, inspiring documentaries, music, and even films,
turning the 52-hertz whale into a symbol of isolation and individuality._
↳Scientifically, however, it remains an open question whether the whale
is truly isolated or simply communicating in ways we do not yet fully_
↳understand, reminding us how much of the ocean-and the lives within it-still
remains a mystery."""

```

```
print("\n===== ORIGINAL TEXT =====\n")
print(text)
```

```
===== ORIGINAL TEXT =====
```

The 52-hertz whale is one of the most fascinating and mysterious creatures ever detected in the oceans, often referred to as the "loneliest whale in the world." This whale was first identified in 1989 by researchers from the Woods Hole Oceanographic Institution while analyzing underwater acoustic data collected by the U.S. Navy. What made this whale extraordinary was the frequency of its call-52 hertz-which is much higher than the typical vocalizations of other large whale species. For example, blue whale usually communicate at frequencies between 10 and 39 hertz, while fin whale calls are typically around 20 hertz. Because sound travels differently underwater, whales rely heavily on low-frequency calls to communicate across vast distances, sometimes hundreds of kilometers. The 52-hertz whale's unusual pitch suggests that other whales may not be able to hear or recognize its calls, which is why it has been popularly described as "lonely," though this interpretation is more poetic than scientifically confirmed. Over the years, the whale has been tracked moving across the North Pacific Ocean, following migration patterns somewhat similar to other baleen whales, yet always singing at its unique frequency. Scientists are still unsure what species it belongs to; some hypothesize it could be a hybrid between different whale species, while others think it might be an individual with a physical anomaly that affects its vocalization system. Despite its uniqueness, the whale does not appear to be distressed-it continues to travel, feed, and vocalize consistently, suggesting it is surviving normally. The idea of its loneliness has captured global imagination, inspiring documentaries, music, and even films, turning the 52-hertz whale into a symbol of isolation and individuality. Scientifically, however, it remains an open question whether the whale is truly isolated or simply communicating in ways we do not yet fully understand, reminding us how much of the ocean-and the lives within it-still remains a mystery.

```
[11]: # =====
# 1. TOKENIZATION FUNCTIONS
# =====

def whitespace_tokenization(text):
    tokenizer = WhitespaceTokenizer()
    return tokenizer.tokenize(text)

def punctuation_tokenization(text):
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```

tokens = []
for word in text.split():
    clean_word = word.strip(string.punctuation)
    if clean_word:
        tokens.append(clean_word)
return tokens

def treebank_tokenization(text):
    tokenizer = TreebankWordTokenizer()
    return tokenizer.tokenize(text)

def tweet_tokenization(text):
    tokenizer = TweetTokenizer()
    return tokenizer.tokenize(text)

def mwe_tokenization(text):
    # Define Multi-Word Expressions
    mwe = MWETokenizer([
        ('Natural', 'Language'),
        ('New', 'York')
    ])
    tokens = word_tokenize(text)
    return mwe.tokenize(tokens)

# -----
# RUN TOKENIZATION
# -----

print("\n===== TOKENIZATION =====\n")

ws_tokens = whitespace_tokenization(text)
print("1. Whitespace Tokens:\n", ws_tokens)

punct_tokens = punctuation_tokenization(text)
print("\n2. Punctuation-based Tokens:\n", punct_tokens)

tree_tokens = treebank_tokenization(text)
print("\n3. Treebank Tokens:\n", tree_tokens)

tweet_tokens = tweet_tokenization(text)
print("\n4. Tweet Tokens:\n", tweet_tokens)

mwe_tokens = mwe_tokenization(text)
print("\n5. MWE Tokens:\n", mwe_tokens)

```

===== TOKENIZATION =====

1. Whitespace Tokens:

['The', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascinating', 'and', 'mysterious', 'creatures', 'ever', 'detected', 'in', 'the', 'oceans', 'often', 'referred', 'to', 'as', 'the', '"loneliest', 'whale', 'in', 'the', 'world."', 'This', 'whale', 'was', 'first', 'identified', 'in', '1989', 'by', 'researchers', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution', 'while', 'analyzing', 'underwater', 'acoustic', 'data', 'collected', 'by', 'the', 'U.S.', 'Navy.', 'What', 'made', 'this', 'whale', 'extraordinary', 'was', 'the', 'frequency', 'of', 'its', 'call-52', 'hertz-which', 'is', 'much', 'higher', 'than', 'the', 'typical', 'vocalizations', 'of', 'other', 'large', 'whale', 'species.', 'For', 'example,', 'blue', 'whale', 'usually', 'communicate', 'at', 'frequencies', 'between', '10', 'and', '39', 'hertz,', 'while', 'fin', 'whale', 'calls', 'are', 'typically', 'around', '20', 'hertz.', 'Because', 'sound', 'travels', 'differently', 'underwater,', 'whales', 'rely', 'heavily', 'on', 'low-frequency', 'calls', 'to', 'communicate', 'across', 'vast', 'distances,', 'sometimes', 'hundreds', 'of', 'kilometers.', 'The', '52-hertz', 'whale's', 'unusual', 'pitch', 'suggests', 'that', 'other', 'whales', 'may', 'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'its', 'calls,', 'which', 'is', 'why', 'it', 'has', 'been', 'popularly', 'described', 'as', '"lonely,"', 'though', 'this', 'interpretation', 'is', 'more', 'poetic', 'than', 'scientifically', 'confirmed.', 'Over', 'the', 'years,', 'the', 'whale', 'has', 'been', 'tracked', 'moving', 'across', 'the', 'North', 'Pacific', 'Ocean,', 'following', 'migration', 'patterns', 'somewhat', 'similar', 'to', 'other', 'baleen', 'whales,', 'yet', 'always', 'singing', 'at', 'its', 'unique', 'frequency.', 'Scientists', 'are', 'still', 'unsure', 'what', 'species', 'it', 'belongs', 'to;', 'some', 'hypothesize', 'it', 'could', 'be', 'a', 'hybrid', 'between', 'different', 'whale', 'species,', 'while', 'others', 'think', 'it', 'might', 'be', 'an', 'individual', 'with', 'a', 'physical', 'anomaly', 'that', 'affects', 'its', 'vocalization', 'system.', 'Despite', 'its', 'uniqueness,', 'the', 'whale', 'does', 'not', 'appear', 'to', 'be', 'distressed-it', 'continues', 'to', 'travel,', 'feed,', 'and', 'vocalize', 'consistently,', 'suggesting', 'it', 'is', 'surviving', 'normally.', 'The', 'idea', 'of', 'its', 'loneliness', 'has', 'captured', 'global', 'imagination,', 'inspiring', 'documentaries,', 'music,', 'and', 'even', 'films,', 'turning', 'the', '52-hertz', 'whale', 'into', 'a', 'symbol', 'of', 'isolation', 'and', 'individuality.', 'Scientifically,', 'however,', 'it', 'remains', 'an', 'open', 'question', 'whether', 'the', 'whale', 'is', 'truly', 'isolated', 'or', 'simply', 'communicating', 'in', 'ways', 'we', 'do', 'not', 'yet', 'fully', 'understand,', 'reminding', 'us', 'how', 'much', 'of', 'the', 'ocean-and', 'the', 'lives', 'within', 'it-still', 'remains', 'a', 'mystery.']}

2. Punctuation-based Tokens:

['The', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascinating', 'and', 'mysterious', 'creatures', 'ever', 'detected', 'in', 'the', 'oceans', 'often', 'referred', 'to', 'as', 'the', '"loneliest', 'whale', 'in', 'the', 'world."', 'This', 'whale', 'was', 'first', 'identified', 'in', '1989', 'by', 'researchers', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution',

'while', 'analyzing', 'underwater', 'acoustic', 'data', 'collected', 'by', 'the', 'U.S.', 'Navy', 'What', 'made', 'this', 'whale', 'extraordinary', 'was', 'the', 'frequency', 'of', 'its', 'call-52', 'hertz-which', 'is', 'much', 'higher', 'than', 'the', 'typical', 'vocalizations', 'of', 'other', 'large', 'whale', 'species', 'For', 'example', 'blue', 'whale', 'usually', 'communicate', 'at', 'frequencies', 'between', '10', 'and', '39', 'hertz', 'while', 'fin', 'whale', 'calls', 'are', 'typically', 'around', '20', 'hertz', 'Because', 'sound', 'travels', 'differently', 'underwater', 'whales', 'rely', 'heavily', 'on', 'low-frequency', 'calls', 'to', 'communicate', 'across', 'vast', 'distances', 'sometimes', 'hundreds', 'of', 'kilometers', 'The', '52-hertz', 'whale's', 'unusual', 'pitch', 'suggests', 'that', 'other', 'whales', 'may', 'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'its', 'calls', 'which', 'is', 'why', 'it', 'has', 'been', 'popularly', 'described', 'as', '"lonely"', 'though', 'this', 'interpretation', 'is', 'more', 'poetic', 'than', 'scientifically', 'confirmed', 'Over', 'the', 'years', 'the', 'whale', 'has', 'been', 'tracked', 'moving', 'across', 'the', 'North', 'Pacific', 'Ocean', 'following', 'migration', 'patterns', 'somewhat', 'similar', 'to', 'other', 'baleen', 'whales', 'yet', 'always', 'singing', 'at', 'its', 'unique', 'frequency', 'Scientists', 'are', 'still', 'unsure', 'what', 'species', 'it', 'belongs', 'to', 'some', 'hypothesize', 'it', 'could', 'be', 'a', 'hybrid', 'between', 'different', 'whale', 'species', 'while', 'others', 'think', 'it', 'might', 'be', 'an', 'individual', 'with', 'a', 'physical', 'anomaly', 'that', 'affects', 'its', 'vocalization', 'system', 'Despite', 'its', 'uniqueness', 'the', 'whale', 'does', 'not', 'appear', 'to', 'be', 'distressed-it', 'continues', 'to', 'travel', 'feed', 'and', 'vocalize', 'consistently', 'suggesting', 'it', 'is', 'surviving', 'normally', 'The', 'idea', 'of', 'its', 'loneliness', 'has', 'captured', 'global', 'imagination', 'inspiring', 'documentaries', 'music', 'and', 'even', 'films', 'turning', 'the', '52-hertz', 'whale', 'into', 'a', 'symbol', 'of', 'isolation', 'and', 'individuality', 'Scientifically', 'however', 'it', 'remains', 'an', 'open', 'question', 'whether', 'the', 'whale', 'is', 'truly', 'isolated', 'or', 'simply', 'communicating', 'in', 'ways', 'we', 'do', 'not', 'yet', 'fully', 'understand', 'reminding', 'us', 'how', 'much', 'of', 'the', 'ocean-and', 'the', 'lives', 'within', 'it-still', 'remains', 'a', 'mystery']

3. Treebank Tokens:

['The', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascinating', 'and', 'mysterious', 'creatures', 'ever', 'detected', 'in', 'the', 'oceans', ',', 'often', 'referred', 'to', 'as', 'the', '"loneliest", 'whale', 'in', 'the', 'world."', 'This', 'whale', 'was', 'first', 'identified', 'in', '1989', 'by', 'researchers', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution', 'while', 'analyzing', 'underwater', 'acoustic', 'data', 'collected', 'by', 'the', 'U.S.', 'Navy.', 'What', 'made', 'this', 'whale', 'extraordinary', 'was', 'the', 'frequency', 'of', 'its', 'call-52', 'hertz-which', 'is', 'much', 'higher', 'than', 'the', 'typical', 'vocalizations', 'of', 'other', 'large', 'whale', 'species.', 'For', 'example', ',', 'blue', 'whale', 'usually', 'communicate', 'at', 'frequencies', 'between', '10', 'and', '39', 'hertz', ',', 'while', 'fin', 'whale', 'calls', 'are', 'typically', 'around', '20', 'hertz.',

'Because', 'sound', 'travels', 'differently', 'underwater', ',', 'whales', 'rely', 'heavily', 'on', 'low-frequency', 'calls', 'to', 'communicate', 'across', 'vast', 'distances', ',', 'sometimes', 'hundreds', 'of', 'kilometers.', 'The', '52-hertz', 'whale's', 'unusual', 'pitch', 'suggests', 'that', 'other', 'whales', 'may', 'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'its', 'calls', ',', 'which', 'is', 'why', 'it', 'has', 'been', 'popularly', 'described', 'as', '"lonely", ',', '"', 'though', 'this', 'interpretation', 'is', 'more', 'poetic', 'than', 'scientifically', 'confirmed.', 'Over', 'the', 'years', ',', 'the', 'whale', 'has', 'been', 'tracked', 'moving', 'across', 'the', 'North', 'Pacific', 'Ocean', ',', 'following', 'migration', 'patterns', 'somewhat', 'similar', 'to', 'other', 'baleen', 'whales', ',', 'yet', 'always', 'singing', 'at', 'its', 'unique', 'frequency.', 'Scientists', 'are', 'still', 'unsure', 'what', 'species', 'it', 'belongs', 'to', ';', 'some', 'hypothesize', 'it', 'could', 'be', 'a', 'hybrid', 'between', 'different', 'whale', 'species', ',', 'while', 'others', 'think', 'it', 'might', 'be', 'an', 'individual', 'with', 'a', 'physical', 'anomaly', 'that', 'affects', 'its', 'vocalization', 'system.', 'Despite', 'its', 'uniqueness', ',', 'the', 'whale', 'does', 'not', 'appear', 'to', 'be', 'distressed-it', 'continues', 'to', 'travel', ',', 'feed', ',', 'and', 'vocalize', 'consistently', ',', 'suggesting', 'it', 'is', 'surviving', 'normally.', 'The', 'idea', 'of', 'its', 'loneliness', 'has', 'captured', 'global', 'imagination', ',', 'inspiring', 'documentaries', ',', 'music', ',', 'and', 'even', 'films', ',', 'turning', 'the', '52-hertz', 'whale', 'into', 'a', 'symbol', 'of', 'isolation', 'and', 'individuality.', 'Scientifically', ',', 'however', ',', 'it', 'remains', 'an', 'open', 'question', 'whether', 'the', 'whale', 'is', 'truly', 'isolated', 'or', 'simply', 'communicating', 'in', 'ways', 'we', 'do', 'not', 'yet', 'fully', 'understand', ',', 'reminding', 'us', 'how', 'much', 'of', 'the', 'ocean-and', 'the', 'lives', 'within', 'it-still', 'remains', 'a', 'mystery', '.']

4. Tweet Tokens:

['The', '52', '-', 'hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascinating', 'and', 'mysterious', 'creatures', 'ever', 'detected', 'in', 'the', 'oceans', ',', 'often', 'referred', 'to', 'as', 'the', '"', 'loneliest', 'whale', 'in', 'the', 'world', '.', '"', 'This', 'whale', 'was', 'first', 'identified', 'in', '1989', 'by', 'researchers', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution', 'while', 'analyzing', 'underwater', 'acoustic', 'data', 'collected', 'by', 'the', 'U', '.', 'S', '.', 'Navy', '.', 'What', 'made', 'this', 'whale', 'extraordinary', 'was', 'the', 'frequency', 'of', 'its', 'call', '-', '52', 'hertz', '-', 'which', 'is', 'much', 'higher', 'than', 'the', 'typical', 'vocalizations', 'of', 'other', 'large', 'whale', 'species', '.', 'For', 'example', ',', 'blue', 'whale', 'usually', 'communicate', 'at', 'frequencies', 'between', '10', 'and', '39', 'hertz', ',', 'while', 'fin', 'whale', 'calls', 'are', 'typically', 'around', '20', 'hertz', '.', 'Because', 'sound', 'travels', 'differently', 'underwater', ',', 'whales', 'rely', 'heavily', 'on', 'low-frequency', 'calls', 'to', 'communicate', 'across', 'vast', 'distances', ',', 'sometimes', 'hundreds', 'of', 'kilometers', '.', 'The', '52', '-', 'hertz', 'whale', 's', 'unusual', 'pitch', 'suggests',

'that', 'other', 'whales', 'may', 'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'its', 'calls', ',', 'which', 'is', 'why', 'it', 'has', 'been', 'popularly', 'described', 'as', '""', 'lonely', ',', '""', 'though', 'this', 'interpretation', 'is', 'more', 'poetic', 'than', 'scientifically', 'confirmed', '.,', 'Over', 'the', 'years', ',', 'the', 'whale', 'has', 'been', 'tracked', 'moving', 'across', 'the', 'North', 'Pacific', 'Ocean', ',', 'following', 'migration', 'patterns', 'somewhat', 'similar', 'to', 'other', 'baleen', 'whales', ',', 'yet', 'always', 'singing', 'at', 'its', 'unique', 'frequency', '.,', 'Scientists', 'are', 'still', 'unsure', 'what', 'species', 'it', 'belongs', 'to', ';;', 'some', 'hypothesize', 'it', 'could', 'be', 'a', 'hybrid', 'between', 'different', 'whale', 'species', ',', 'while', 'others', 'think', 'it', 'might', 'be', 'an', 'individual', 'with', 'a', 'physical', 'anomaly', 'that', 'affects', 'its', 'vocalization', 'system', '.,', 'Despite', 'its', 'uniqueness', ',', 'the', 'whale', 'does', 'not', 'appear', 'to', 'be', 'distressed', '- ', 'it', 'continues', 'to', 'travel', ',', 'feed', ',', 'and', 'vocalize', 'consistently', ',', 'suggesting', 'it', 'is', 'surviving', 'normally', '.,', 'The', 'idea', 'of', 'its', 'loneliness', 'has', 'captured', 'global', 'imagination', ',', 'inspiring', 'documentaries', ',', 'music', ',', 'and', 'even', 'films', ',', 'turning', 'the', '52', '- ', 'hertz', 'whale', 'into', 'a', 'symbol', 'of', 'isolation', 'and', 'individuality', '.,', 'Scientifically', ',', 'however', ',', 'it', 'remains', 'an', 'open', 'question', 'whether', 'the', 'whale', 'is', 'truly', 'isolated', 'or', 'simply', 'communicating', 'in', 'ways', 'we', 'do', 'not', 'yet', 'fully', 'understand', ',', 'reminding', 'us', 'how', 'much', 'of', 'the', 'ocean', '- ', 'and', 'the', 'lives', 'within', 'it', '- ', 'still', 'remains', 'a', 'mystery', '.]'

5. MWE Tokens:

['The', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascinating', 'and', 'mysterious', 'creatures', 'ever', 'detected', 'in', 'the', 'oceans', ',', 'often', 'referred', 'to', 'as', 'the', '""', 'loneliest', 'whale', 'in', 'the', 'world.', '""', 'This', 'whale', 'was', 'first', 'identified', 'in', '1989', 'by', 'researchers', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution', 'while', 'analyzing', 'underwater', 'acoustic', 'data', 'collected', 'by', 'the', 'U.S.', 'Navy', '.,', 'What', 'made', 'this', 'whale', 'extraordinary', 'was', 'the', 'frequency', 'of', 'its', 'call-52', 'hertz-which', 'is', 'much', 'higher', 'than', 'the', 'typical', 'vocalizations', 'of', 'other', 'large', 'whale', 'species', '.,', 'For', 'example', ',', 'blue', 'whale', 'usually', 'communicate', 'at', 'frequencies', 'between', '10', 'and', '39', 'hertz', ',', 'while', 'fin', 'whale', 'calls', 'are', 'typically', 'around', '20', 'hertz', '.,', 'Because', 'sound', 'travels', 'differently', 'underwater', ',', 'whales', 'rely', 'heavily', 'on', 'low-frequency', 'calls', 'to', 'communicate', 'across', 'vast', 'distances', ',', 'sometimes', 'hundreds', 'of', 'kilometers', '.,', 'The', '52-hertz', 'whale', 's', 'unusual', 'pitch', 'suggests', 'that', 'other', 'whales', 'may', 'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'its', 'calls', ',', 'which', 'is', 'why', 'it', 'has', 'been', 'popularly', 'described', 'as', '""', 'lonely', ',', '""', 'though', 'this', 'interpretation', 'is', 'more', 'poetic', 'than', 'scientifically', 'confirmed', '.,', 'Over', 'the', 'years', ',', 'the',

'whale', 'has', 'been', 'tracked', 'moving', 'across', 'the', 'North', 'Pacific', 'Ocean', ',', 'following', 'migration', 'patterns', 'somewhat', 'similar', 'to', 'other', 'baleen', 'whales', ',', 'yet', 'always', 'singing', 'at', 'its', 'unique', 'frequency', '.', 'Scientists', 'are', 'still', 'unsure', 'what', 'species', 'it', 'belongs', 'to', ';', 'some', 'hypothesize', 'it', 'could', 'be', 'a', 'hybrid', 'between', 'different', 'whale', 'species', ',', 'while', 'others', 'think', 'it', 'might', 'be', 'an', 'individual', 'with', 'a', 'physical', 'anomaly', 'that', 'affects', 'its', 'vocalization', 'system', '.', 'Despite', 'its', 'uniqueness', ',', 'the', 'whale', 'does', 'not', 'appear', 'to', 'be', 'distressed-it', 'continues', 'to', 'travel', ',', 'feed', ',', 'and', 'vocalize', 'consistently', ',', 'suggesting', 'it', 'is', 'surviving', 'normally', '.', 'The', 'idea', 'of', 'its', 'loneliness', 'has', 'captured', 'global', 'imagination', ',', 'inspiring', 'documentaries', ',', 'music', ',', 'and', 'even', 'films', ',', 'turning', 'the', '52-hertz', 'whale', 'into', 'a', 'symbol', 'of', 'isolation', 'and', 'individuality', '.', 'Scientifically', ',', 'however', ',', 'it', 'remains', 'an', 'open', 'question', 'whether', 'the', 'whale', 'is', 'truly', 'isolated', 'or', 'simply', 'communicating', 'in', 'ways', 'we', 'do', 'not', 'yet', 'fully', 'understand', ',', 'reminding', 'us', 'how', 'much', 'of', 'the', 'ocean-and', 'the', 'lives', 'within', 'it-still', 'remains', 'a', 'mystery', '.']

```
[12]: # =====
# 2. STEMMING
# =====

def porter_stemming(tokens):
    stemmer = PorterStemmer()
    return [stemmer.stem(word) for word in tokens]

def snowball_stemming(tokens):
    stemmer = SnowballStemmer("english")
    return [stemmer.stem(word) for word in tokens]

print("\n===== STEMMING =====\n")

porter_result = porter_stemming(tree_tokens)
print("6. Porter Stemmer Output:\n", porter_result)

snowball_result = snowball_stemming(tree_tokens)
print("\n7. Snowball Stemmer Output:\n", snowball_result)
```

===== STEMMING =====

6. Porter Stemmer Output:

['the', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascin', 'and', 'mysteri', 'creatur', 'ever', 'detect', 'in', 'the', 'ocean', ',', 'often', 'refer', 'to', 'as', 'the', '"loneliest', 'whale', 'in', 'the', 'world."',

'thi', 'whale', 'wa', 'first', 'identifi', 'in', '1989', 'by', 'research',
 'from', 'the', 'wood', 'hole', 'oceanograph', 'institut', 'while', 'analyz',
 'underwat', 'acoust', 'data', 'collect', 'by', 'the', 'u.s.', 'navy.', 'what',
 'made', 'thi', 'whale', 'extraordinari', 'wa', 'the', 'frequenc', 'of', 'it',
 'call-52', 'hertz-which', 'is', 'much', 'higher', 'than', 'the', 'typic',
 'vocal', 'of', 'other', 'larg', 'whale', 'species.', 'for', 'exempl', ',',
 'blue', 'whale', 'usual', 'commun', 'at', 'frequenc', 'between', '10', 'and',
 '39', 'hertz', ',', 'while', 'fin', 'whale', 'call', 'are', 'typic', 'around',
 '20', 'hertz.', 'becaus', 'sound', 'travel', 'differ', 'underwat', ',', 'whale',
 'reli', 'heavili', 'on', 'low-frequ', 'call', 'to', 'commun', 'across', 'vast',
 'distan', ',', 'sometim', 'hundr', 'of', 'kilometers.', 'the', '52-hertz',
 'whale', 'unusu', 'pitch', 'suggest', 'that', 'other', 'whale', 'may', 'not',
 'be', 'abl', 'to', 'hear', 'or', 'recogn', 'it', 'call', ',', 'which', 'is',
 'whi', 'it', 'ha', 'been', 'popularli', 'describ', 'as', '"lone', ',', '"',
 'though', 'thi', 'interpret', 'is', 'more', 'poetic', 'than', 'scientif',
 'confirmed.', 'over', 'the', 'year', ',', 'the', 'whale', 'ha', 'been', 'track',
 'move', 'across', 'the', 'north', 'pacif', 'ocean', ',', 'follow', 'migrat',
 'pattern', 'somewhat', 'similar', 'to', 'other', 'baleen', 'whale', ',', 'yet',
 'alway', 'sing', 'at', 'it', 'uniqu', 'frequency.', 'scientist', 'are', 'still',
 'unsur', 'what', 'speci', 'it', 'belong', 'to', ';', 'some', 'hypothes', 'it',
 'could', 'be', 'a', 'hybrid', 'between', 'differ', 'whale', 'speci', ',',
 'while', 'other', 'think', 'it', 'might', 'be', 'an', 'individu', 'with', 'a',
 'physic', 'anomali', 'that', 'affect', 'it', 'vocal', 'system.', 'despit', 'it',
 'uniqu', ',', 'the', 'whale', 'doe', 'not', 'appear', 'to', 'be',
 'distressed-it', 'continu', 'to', 'travel', ',', 'feed', ',', 'and', 'vocal',
 'consist', ',', 'suggest', 'it', 'is', 'surviv', 'normally.', 'the', 'idea',
 'of', 'it', 'loneli', 'ha', 'captur', 'global', 'imagin', ',', 'inspir',
 'documentari', ',', 'music', ',', 'and', 'even', 'film', ',', 'turn', 'the',
 '52-hertz', 'whale', 'into', 'a', 'symbol', 'of', 'isol', 'and',
 'individuality.', 'scientif', ',', 'howev', ',', 'it', 'remain', 'an', 'open',
 'question', 'whether', 'the', 'whale', 'is', 'truli', 'isol', 'or', 'simpli',
 'commun', 'in', 'way', 'we', 'do', 'not', 'yet', 'fulli', 'understand', ',',
 'remind', 'us', 'how', 'much', 'of', 'the', 'ocean-and', 'the', 'live',
 'within', 'it-stil', 'remain', 'a', 'mysteri', '.']

7. Snowball Stemmer Output:

['the', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascin', 'and',
 'mysteri', 'creatur', 'ever', 'detect', 'in', 'the', 'ocean', ',', 'often',
 'refer', 'to', 'as', 'the', '"loneliest', 'whale', 'in', 'the', 'world."',
 'this', 'whale', 'was', 'first', 'identifi', 'in', '1989', 'by', 'research',
 'from', 'the', 'wood', 'hole', 'oceanograph', 'institut', 'while', 'analyz',
 'underwat', 'acoust', 'data', 'collect', 'by', 'the', 'u.s.', 'navy.', 'what',
 'made', 'this', 'whale', 'extraordinari', 'was', 'the', 'frequenc', 'of', 'it',
 'call-52', 'hertz-which', 'is', 'much', 'higher', 'than', 'the', 'typic',
 'vocal', 'of', 'other', 'larg', 'whale', 'species.', 'for', 'exempl', ',',
 'blue', 'whale', 'usual', 'communic', 'at', 'frequenc', 'between', '10', 'and',
 '39', 'hertz', ',', 'while', 'fin', 'whale', 'call', 'are', 'typic', 'around',
 '20', 'hertz.', 'becaus', 'sound', 'travel', 'differ', 'underwat', ',', 'whale',

'reli', 'heavili', 'on', 'low-frequ', 'call', 'to', 'communic', 'across',
 'vast', 'distanc', ',', 'sometim', 'hundr', 'of', 'kilometers.', 'the',
 '52-hertz', 'whale', 'unusu', 'pitch', 'suggest', 'that', 'other', 'whale',
 'may', 'not', 'be', 'abl', 'to', 'hear', 'or', 'recogn', 'it', 'call', ',',
 'which', 'is', 'whi', 'it', 'has', 'been', 'popular', 'describ', 'as', '"lone',
 ',', '"', 'though', 'this', 'interpret', 'is', 'more', 'poetic', 'than',
 'scientif', 'confirmed.', 'over', 'the', 'year', ',', 'the', 'whale', 'has',
 'been', 'track', 'move', 'across', 'the', 'north', 'pacif', 'ocean', ',',
 'follow', 'migrat', 'pattern', 'somewhat', 'similar', 'to', 'other', 'baleen',
 'whale', ',', 'yet', 'alway', 'sing', 'at', 'it', 'uniqu', 'frequency.',
 'scientist', 'are', 'still', 'unsur', 'what', 'speci', 'it', 'belong', 'to',
 ';', 'some', 'hypothes', 'it', 'could', 'be', 'a', 'hybrid', 'between',
 'differ', 'whale', 'speci', ',', 'while', 'other', 'think', 'it', 'might', 'be',
 'an', 'individu', 'with', 'a', 'physic', 'anomali', 'that', 'affect', 'it',
 'vocal', 'system.', 'despit', 'it', 'uniqu', ',', 'the', 'whale', 'doe', 'not',
 'appear', 'to', 'be', 'distressed-it', 'continu', 'to', 'travel', ',', 'feed',
 ',', 'and', 'vocal', 'consist', ',', 'suggest', 'it', 'is', 'surviv',
 'normally.', 'the', 'idea', 'of', 'it', 'loneli', 'has', 'captur', 'global',
 'imagin', ',', 'inspir', 'documentari', ',', 'music', ',', 'and', 'even',
 'film', ',', 'turn', 'the', '52-hertz', 'whale', 'into', 'a', 'symbol', 'of',
 'isol', 'and', 'individuality.', 'scientif', ',', 'howev', ',', 'it', 'remain',
 'an', 'open', 'question', 'whether', 'the', 'whale', 'is', 'truli', 'isol',
 'or', 'simpli', 'communic', 'in', 'way', 'we', 'do', 'not', 'yet', 'fulli',
 'understand', ',', 'remind', 'us', 'how', 'much', 'of', 'the', 'ocean-and',
 'the', 'live', 'within', 'it-stil', 'remain', 'a', 'mysteri', '.']

```
[15]: # =====
# 3. LEMMATIZATION
# =====

lemmatizer = WordNetLemmatizer()

def simple_lemmatization(tokens):
    return [lemmatizer.lemmatize(word) for word in tokens]

# POS tagging helper
def get_wordnet_pos(tag):
    if tag.startswith('J'):
        return 'a' # adjective
    elif tag.startswith('V'):
        return 'v' # verb
    elif tag.startswith('N'):
        return 'n' # noun
    elif tag.startswith('R'):
        return 'r' # adverb
    else:
        return 'n'
```

```

def pos_lemmatization(tokens):
    pos_tags = nltk.pos_tag(tokens)
    return [
        lemmatizer.lemmatize(word, get_wordnet_pos(tag))
        for word, tag in pos_tags
    ]

print("\n===== LEMMATIZATION =====\n")

simple_lemmas = simple_lemmatization(tree_tokens)
print("8. Simple Lemmatization:\n", simple_lemmas)

pos_lemmas = pos_lemmatization(tree_tokens)
print("\n9. POS-based Lemmatization:\n", pos_lemmas)

```

===== LEMMATIZATION =====

8. Simple Lemmatization:

```

['The', '52-hertz', 'whale', 'is', 'one', 'of', 'the', 'most', 'fascinating',
'and', 'mysterious', 'creature', 'ever', 'detected', 'in', 'the', 'ocean', ',',
'often', 'referred', 'to', 'a', 'the', '"loneliest', 'whale', 'in', 'the',
'world."', 'This', 'whale', 'was', 'first', 'identified', 'in', '1989', 'by',
'researcher', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution',
'while', 'analyzing', 'underwater', 'acoustic', 'data', 'collected', 'by',
'the', 'U.S.', 'Navy.', 'What', 'made', 'this', 'whale', 'extraordinary', 'was',
'the', 'frequency', 'of', 'it', 'call-52', 'hertz-which', 'is', 'much',
'higher', 'than', 'the', 'typical', 'vocalization', 'of', 'other', 'large',
'whale', 'species.', 'For', 'example', ',', 'blue', 'whale', 'usually',
'communicate', 'at', 'frequency', 'between', '10', 'and', '39', 'hertz', ',',
'while', 'fin', 'whale', 'call', 'are', 'typically', 'around', '20', 'hertz.',
'Because', 'sound', 'travel', 'differently', 'underwater', ',', 'whale', 'rely',
'heavily', 'on', 'low-frequency', 'call', 'to', 'communicate', 'across', 'vast',
'distance', ',', 'sometimes', 'hundred', 'of', 'kilometers.', 'The', '52-hertz',
'whale's', 'unusual', 'pitch', 'suggests', 'that', 'other', 'whale', 'may',
'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'it', 'call', ',',
'which', 'is', 'why', 'it', 'has', 'been', 'popularly', 'described', 'as',
'"lonely', ',', '"', 'though', 'this', 'interpretation', 'is', 'more', 'poetic',
'than', 'scientifically', 'confirmed.', 'Over', 'the', 'year', ',', 'the',
'whale', 'has', 'been', 'tracked', 'moving', 'across', 'the', 'North', 'Pacific',
'Ocean', ',', 'following', 'migration', 'pattern', 'somewhat', 'similar', 'to',
'other', 'baleen', 'whale', ',', 'yet', 'always', 'singing', 'at', 'it',
'unique', 'frequency.', 'Scientists', 'are', 'still', 'unsure', 'what',
'species', 'it', 'belongs', 'to', ';', 'some', 'hypothesize', 'it', 'could',
'be', 'a', 'hybrid', 'between', 'different', 'whale', 'species', ',', 'while',
'others', 'think', 'it', 'might', 'be', 'an', 'individual', 'with', 'a',

```

'physical', 'anomaly', 'that', 'affect', 'it', 'vocalization', 'system.',
 'Despite', 'it', 'uniqueness', ',', 'the', 'whale', 'doe', 'not', 'appear',
 'to', 'be', 'distressed-it', 'continues', 'to', 'travel', ',', 'feed', ',',
 'and', 'vocalize', 'consistently', ',', 'suggesting', 'it', 'is', 'surviving',
 'normally.', 'The', 'idea', 'of', 'it', 'loneliness', 'ha', 'captured',
 'global', 'imagination', ',', 'inspiring', 'documentary', ',', 'music', ',',
 'and', 'even', 'film', ',', 'turning', 'the', '52-hertz', 'whale', 'into', 'a',
 'symbol', 'of', 'isolation', 'and', 'individuality.', 'Scientifically', ',',
 'however', ',', 'it', 'remains', 'an', 'open', 'question', 'whether', 'the',
 'whale', 'is', 'truly', 'isolated', 'or', 'simply', 'communicating', 'in',
 'way', 'we', 'do', 'not', 'yet', 'fully', 'understand', ',', 'reminding', 'u',
 'how', 'much', 'of', 'the', 'ocean-and', 'the', 'life', 'within', 'it-still',
 'remains', 'a', 'mystery', '.']

9. POS-based Lemmatization:

['The', '52-hertz', 'whale', 'be', 'one', 'of', 'the', 'most', 'fascinating',
 'and', 'mysterious', 'creature', 'ever', 'detect', 'in', 'the', 'ocean', ',',
 'often', 'refer', 'to', 'a', 'the', '"loneliest', 'whale', 'in', 'the',
 'world."', 'This', 'whale', 'be', 'first', 'identify', 'in', '1989', 'by',
 'researcher', 'from', 'the', 'Woods', 'Hole', 'Oceanographic', 'Institution',
 'while', 'analyze', 'underwater', 'acoustic', 'data', 'collect', 'by', 'the',
 'U.S.', 'Navy.', 'What', 'make', 'this', 'whale', 'extraordinary', 'be', 'the',
 'frequency', 'of', 'it', 'call-52', 'hertz-which', 'be', 'much', 'high', 'than',
 'the', 'typical', 'vocalization', 'of', 'other', 'large', 'whale', 'species.',
 'For', 'example', ',', 'blue', 'whale', 'usually', 'communicate', 'at',
 'frequency', 'between', '10', 'and', '39', 'hertz', ',', 'while', 'fin',
 'whale', 'call', 'be', 'typically', 'around', '20', 'hertz.', 'Because',
 'sound', 'travel', 'differently', 'underwater', ',', 'whale', 'rely', 'heavily',
 'on', 'low-frequency', 'call', 'to', 'communicate', 'across', 'vast',
 'distance', ',', 'sometimes', 'hundred', 'of', 'kilometers.', 'The', '52-hertz',
 'whale's', 'unusual', 'pitch', 'suggest', 'that', 'other', 'whale', 'may',
 'not', 'be', 'able', 'to', 'hear', 'or', 'recognize', 'it', 'call', ',',
 'which', 'be', 'why', 'it', 'have', 'be', 'popularly', 'describe', 'a',
 '"lonely', ',', 'though', 'this', 'interpretation', 'be', 'more', 'poetic',
 'than', 'scientifically', 'confirmed.', 'Over', 'the', 'year', ',', 'the',
 'whale', 'have', 'be', 'track', 'move', 'across', 'the', 'North', 'Pacific',
 'Ocean', ',', 'follow', 'migration', 'pattern', 'somewhat', 'similar', 'to',
 'other', 'baleen', 'whale', ',', 'yet', 'always', 'sing', 'at', 'it', 'unique',
 'frequency.', 'Scientists', 'be', 'still', 'unsure', 'what', 'species', 'it',
 'belong', 'to', ';', 'some', 'hypothesize', 'it', 'could', 'be', 'a', 'hybrid',
 'between', 'different', 'whale', 'specie', ',', 'while', 'others', 'think',
 'it', 'might', 'be', 'an', 'individual', 'with', 'a', 'physical', 'anomaly',
 'that', 'affect', 'it', 'vocalization', 'system.', 'Despite', 'it',
 'uniqueness', ',', 'the', 'whale', 'do', 'not', 'appear', 'to', 'be',
 'distressed-it', 'continue', 'to', 'travel', ',', 'feed', ',', 'and',
 'vocalize', 'consistently', ',', 'suggest', 'it', 'be', 'survive', 'normally.',
 'The', 'idea', 'of', 'it', 'loneliness', 'have', 'capture', 'global',
 'imagination', ',', 'inspire', 'documentary', ',', 'music', ',', 'and', 'even',

```
'film', ',', 'turn', 'the', '52-hertz', 'whale', 'into', 'a', 'symbol', 'of',
'isolation', 'and', 'individuality.', 'Scientifically', ',', 'however', ',', 'it', 'remain', 'an', 'open', 'question', 'whether', 'the', 'whale', 'be',
'truly', 'isolated', 'or', 'simply', 'communicate', 'in', 'way', 'we', 'do',
'not', 'yet', 'fully', 'understand', ',', 'remind', 'u', 'how', 'much', 'of',
'the', 'ocean-and', 'the', 'life', 'within', 'it-still', 'remain', 'a',
'mystery', '.']
```

```
[16]: # =====
# 4. COMPARISON TABLE
# =====

print("\n===== COMPARISON TABLE =====\n")

print("{:<15} {:<15} {:<15} {:<15}".format(
    "Original", "Porter", "Snowball", "Lemma"
))

for i in range(min(len(tree_tokens), 10)): # limit for readability
    print("{:<15} {:<15} {:<15} {:<15}".format(
        tree_tokens[i],
        porter_result[i],
        snowball_result[i],
        pos_lemmas[i]
    ))
```

```
===== COMPARISON TABLE =====
```

Original	Porter	Snowball	Lemma
The	the	the	The
52-hertz	52-hertz	52-hertz	52-hertz
whale	whale	whale	whale
is	is	is	be
one	one	one	one
of	of	of	of
the	the	the	the
most	most	most	most
fascinating	fascin	fascin	fascinating
and	and	and	and

```
[17]: # =====
# 5. SUMMARY
# =====

print("\n===== SUMMARY =====\n")
```

```

print(f"Total Tokens (Treebank): {len(tree_tokens)}")
print(f"Unique Tokens: {len(set(tree_tokens))}")
print(f"After Stemming (Porter): {len(set(porter_result))}")
print(f"After Lemmatization: {len(set(pos_lemmas))}")

print("\nPROCESS COMPLETED SUCCESSFULLY ")

import matplotlib.pyplot as plt
from collections import Counter

# =====
# 1. TOKEN FREQUENCY DISTRIBUTION
# =====

freq_dist = Counter(tree_tokens)

words = list(freq_dist.keys())
counts = list(freq_dist.values())

plt.figure()
plt.bar(words, counts)
plt.xticks(rotation=90)
plt.title("Token Frequency Distribution")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

```

===== SUMMARY =====

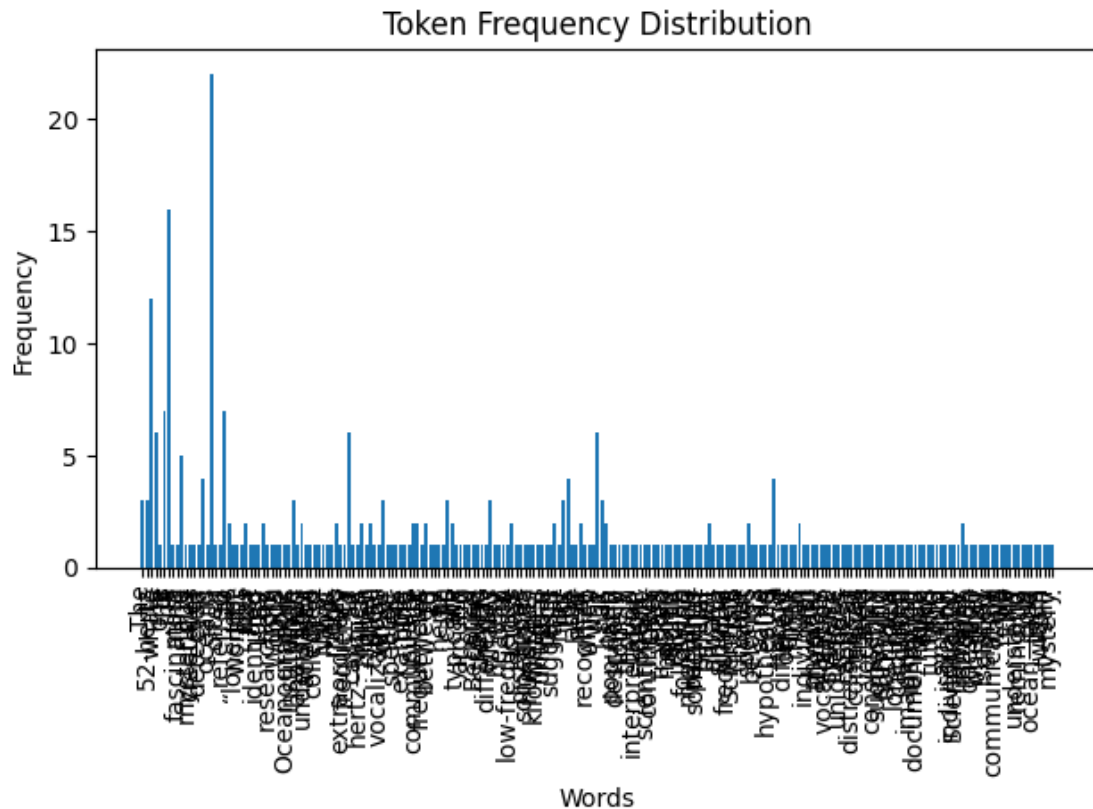
Total Tokens (Treebank): 335

Unique Tokens: 213

After Stemming (Porter): 194

After Lemmatization: 201

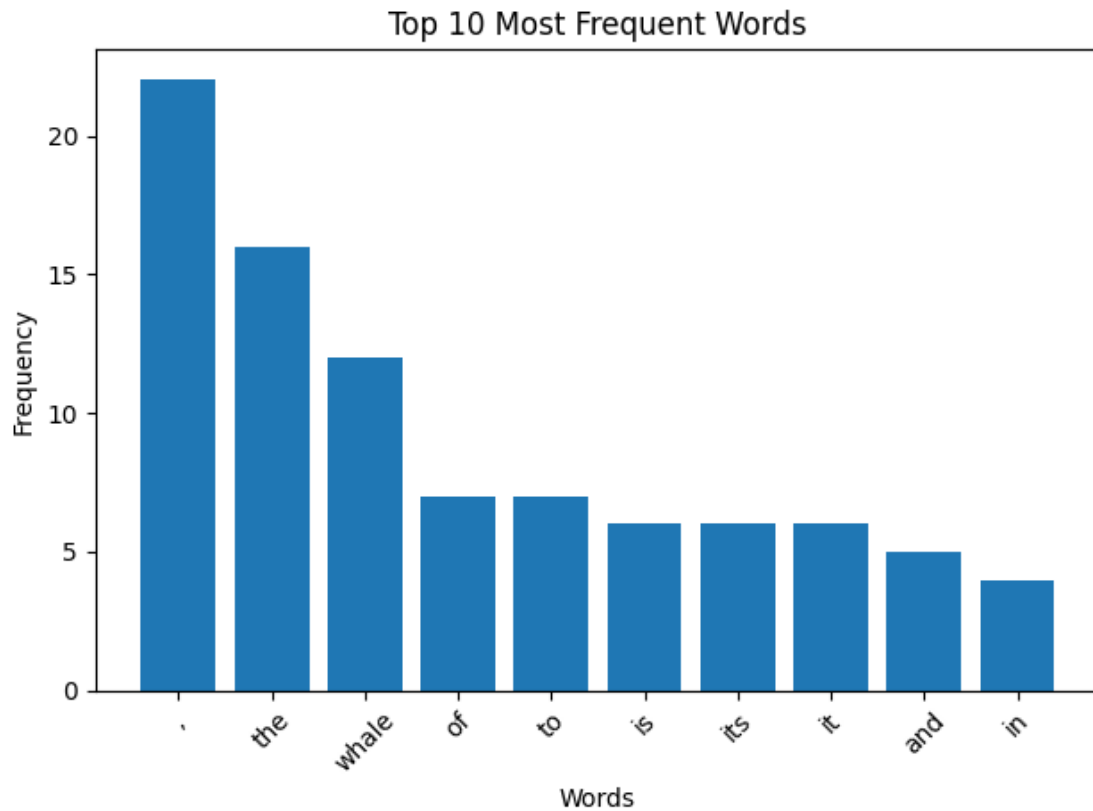
PROCESS COMPLETED SUCCESSFULLY



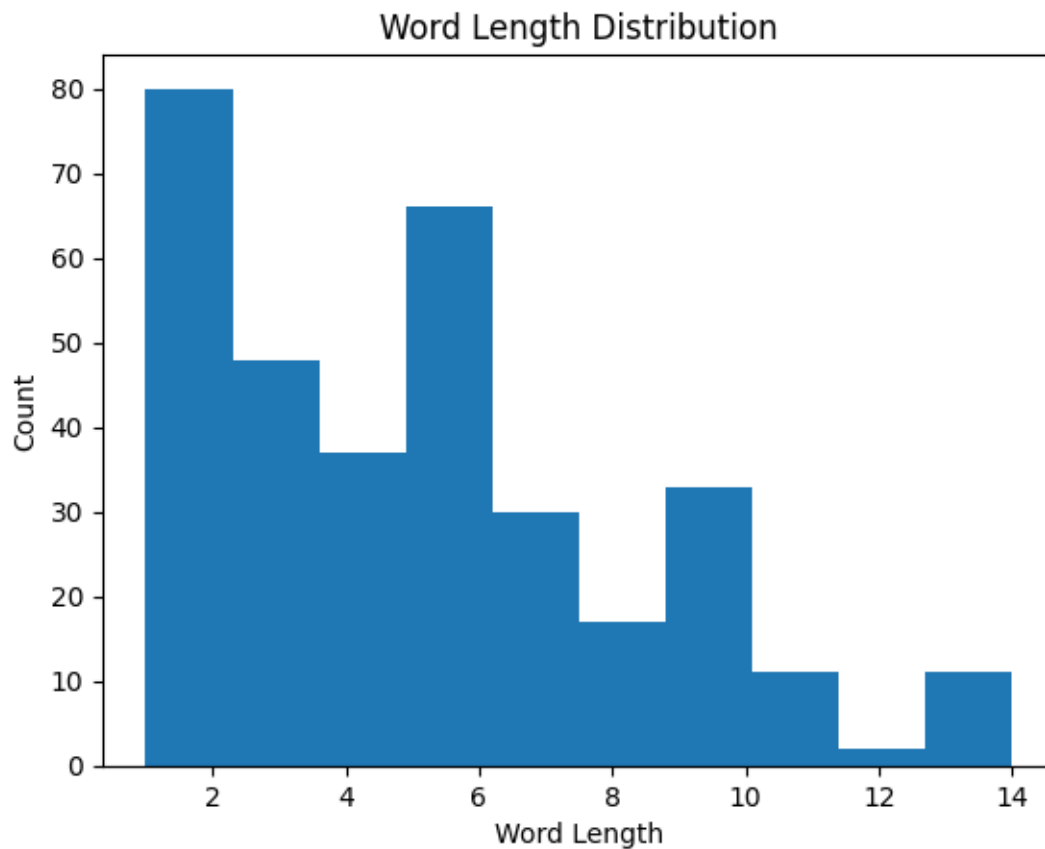
```
[18]: # =====
# 2. TOP 10 MOST COMMON WORDS
# =====

top_words = freq_dist.most_common(10)
top_w, top_c = zip(*top_words)

plt.figure()
plt.bar(top_w, top_c)
plt.title("Top 10 Most Frequent Words")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

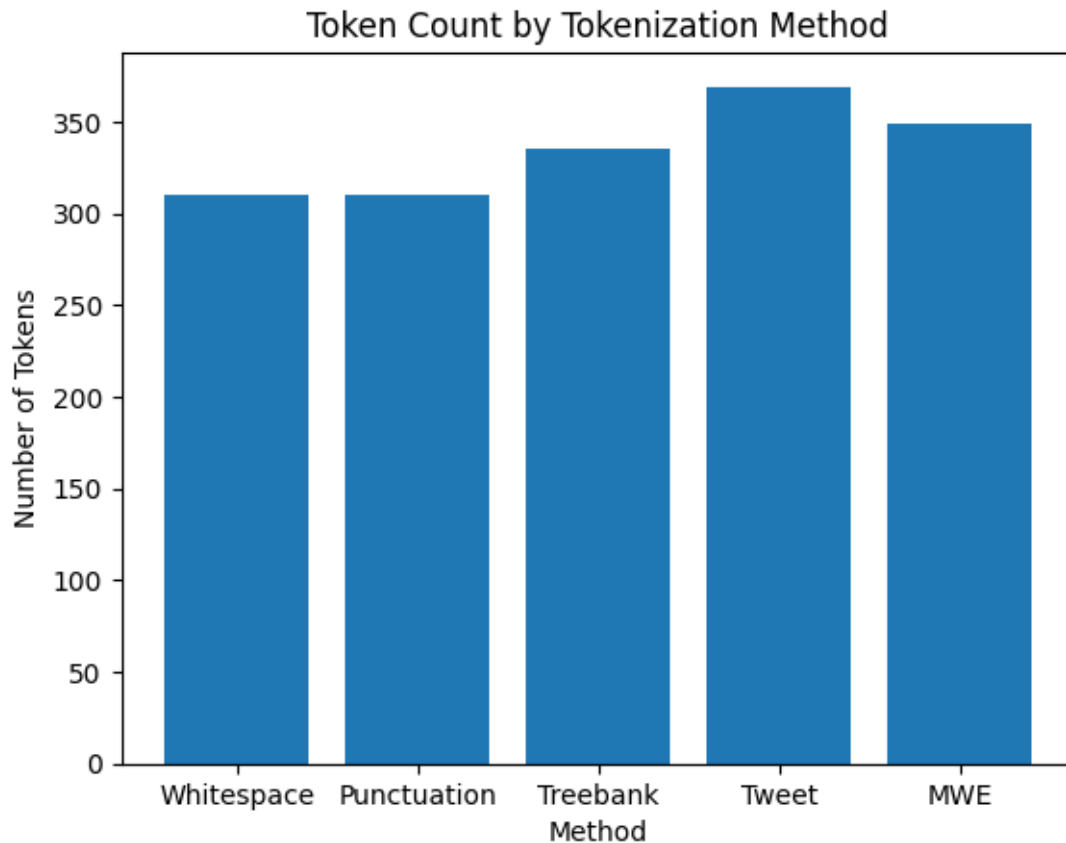
```
[19]: # =====  
# 3. WORD LENGTH DISTRIBUTION  
# =====  
  
word_lengths = [len(word) for word in tree_tokens]  
  
plt.figure()  
plt.hist(word_lengths, bins=10)  
plt.title("Word Length Distribution")  
plt.xlabel("Word Length")  
plt.ylabel("Count")  
plt.show()
```



```
[20]: # =====
# 4. TOKEN COUNT COMPARISON (Different Methods)
# =====

methods = ['Whitespace', 'Punctuation', 'Treebank', 'Tweet', 'MWE']
counts = [
    len(ws_tokens),
    len(punct_tokens),
    len(tree_tokens),
    len(tweet_tokens),
    len(mwe_tokens)
]

plt.figure()
plt.bar(methods, counts)
plt.title("Token Count by Tokenization Method")
plt.xlabel("Method")
plt.ylabel("Number of Tokens")
plt.show()
```

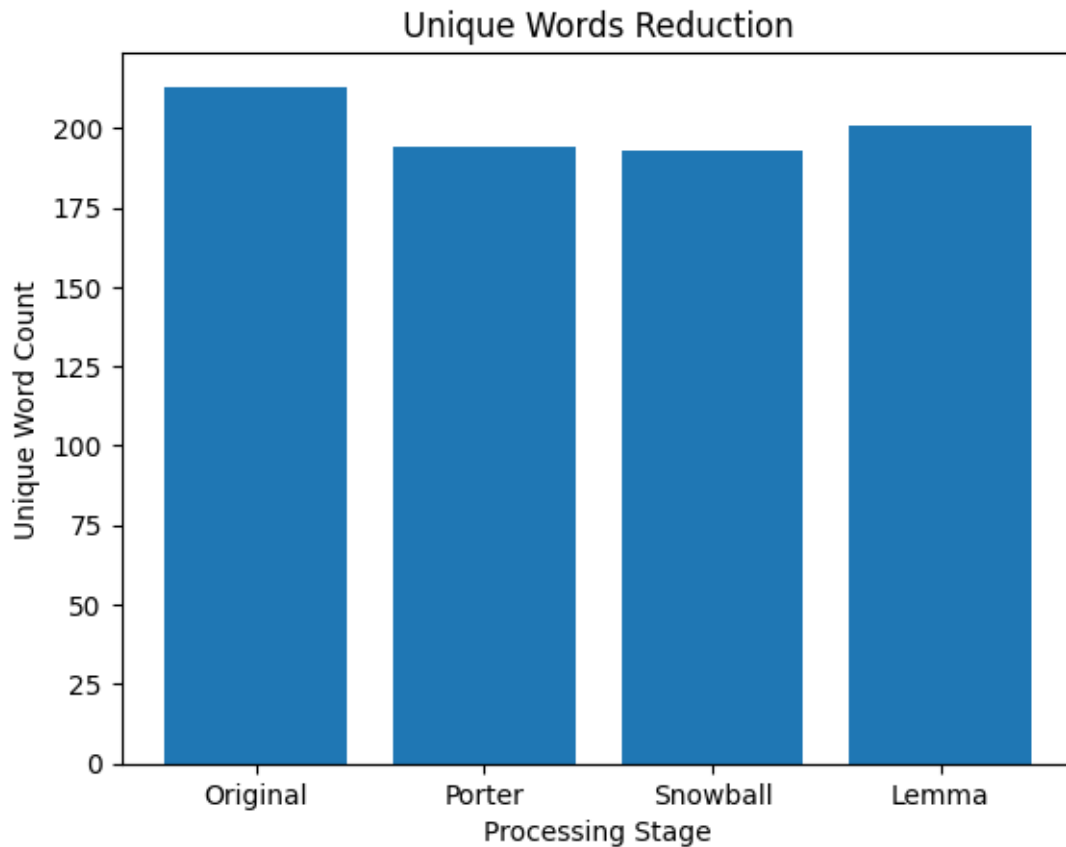


```
[21]: # =====
# 5. UNIQUE WORD REDUCTION
# =====

original_unique = len(set(tree_tokens))
porter_unique = len(set(porter_result))
snowball_unique = len(set(snowball_result))
lemma_unique = len(set(pos_lemmas))

labels = ['Original', 'Porter', 'Snowball', 'Lemma']
values = [original_unique, porter_unique, snowball_unique, lemma_unique]

plt.figure()
plt.bar(labels, values)
plt.title("Unique Words Reduction")
plt.xlabel("Processing Stage")
plt.ylabel("Unique Word Count")
plt.show()
```



```
[22]: # =====
# 6. STEMMING vs LEMMATIZATION COMPARISON
# =====

sample_words = tree_tokens[:10]

porter_sample = porter_result[:10]
snowball_sample = snowball_result[:10]
lemma_sample = pos_lemmas[:10]

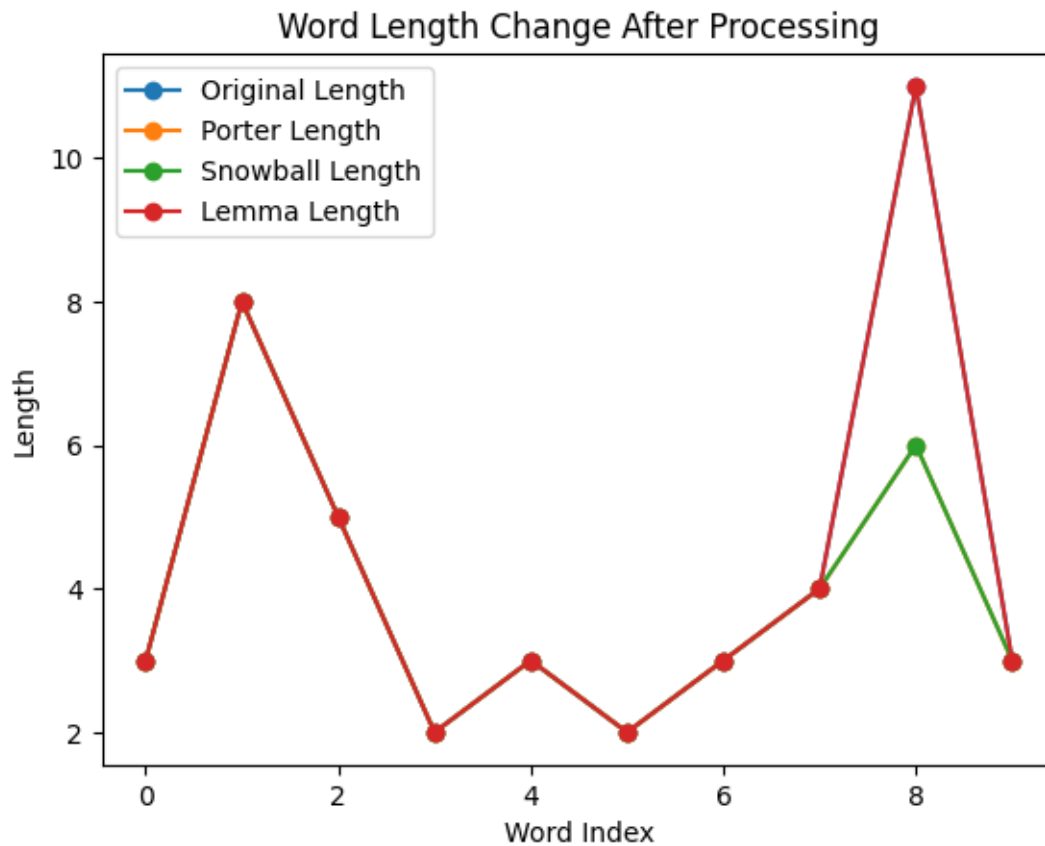
x = range(len(sample_words))

plt.figure()
plt.plot(x, [len(w) for w in sample_words], marker='o', label="Original Length")
plt.plot(x, [len(w) for w in porter_sample], marker='o', label="Porter Length")
plt.plot(x, [len(w) for w in snowball_sample], marker='o', label="Snowball_
↳Length")
plt.plot(x, [len(w) for w in lemma_sample], marker='o', label="Lemma Length")

plt.title("Word Length Change After Processing")
```

```
plt.xlabel("Word Index")
plt.ylabel("Length")
plt.legend()
plt.show()

print("\nVISUALIZATION COMPLETED ")
```



VISUALIZATION COMPLETED

[]: